

Week 6 Finish and restart · paper route

Paper route · GROW Computing · 40 minutes

Name: _____ Date: _____

Week 6 paper route. Finish and restart.

I can create a win response and start a fresh game.

This is a full route. You can finish Week 6 with this booklet, a pen and one counter.

This is not a rehearsal. It is not practice for a computer lesson.

The work you do on these pages is your Week 6 work. It is checked here.

You do not need to do it again on a computer.

Your teacher watches you work. Your teacher writes down what you did.

You can point, say, draw or write. An adult can write your words for you.

You need three things: this booklet, a pen or pencil, one counter.

A counter can be a coin, a button, or a small paper square.

Ask for help when you need it. Help is written down. Help is not a mark against you.

Page 1 is your block writing page. Page 2 is your maze. Page 3 is your testing page.

Point, say, draw or write. Someone may record your exact words.

On the screen route this week is saved as W06_Win.sb3. On the paper route no file is made. Your pages are the record: write your name and the date on every page and put them in your folder. An adult initials the folder. Nobody should ask you for W06_Win.sb3 if you took the paper route.

Write your code

Player · This week's new script. It is the win response and the fresh start. Write this one yourself. This is the grid your teacher watches you fill in. One row has a hexagon box. Write the condition inside the hexagon.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- say [blank]
- say [Made it!]
- say [your own win message]
- go to x [-180] y [-120]
- wait until <>
- touching [Finish]?
- touching [Maze]?
- forever
- if <> then
- say [Made it!] for [2] seconds

Player · Your wall rule from Week 5. You already built this. Write it again here so your whole game is on one page. Check 4 needs this rule to still work after a win.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- forever
- if <> then

- wait until <>
- touching [Maze]?
- touching [Finish]?
- go to x [-180] y [-120]

Player · Your right arrow control from Week 4. Write it again here. It is one of the four key scripts. It must still work after the win message.

Write one block on each line. Start with the hat block, which is done for you.

when [right arrow] key pressed

Blocks you may use this week:

- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Your left arrow control from Week 4. Write it again here. It is one of the four key scripts.

Write one block on each line. Start with the hat block, which is done for you.

when [left arrow] key pressed

Blocks you may use this week:

- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Your up arrow control from Week 4. Write it again here. It is one of the four key scripts.

Write one block on each line. Start with the hat block, which is done for you.

when [up arrow] key pressed

Blocks you may use this week:

- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Your down arrow control from Week 4. Write it again here. It is one of the four key scripts.

Write one block on each line. Start with the hat block, which is done for you.

when [down arrow] key pressed

Blocks you may use this week:

- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Your start script from Week 4. It puts Player on the safe start. Use the same x and y as the go to block in your new win script. All your start blocks use the same numbers.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- go to x [-180] y [-120]
- say [blank]

Player · Challenge only. Do the win and the restart first. The broken win script is printed for you on the challenge page. Read it. Then write the repaired script in this grid. The repaired script must match your grid 1 script.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

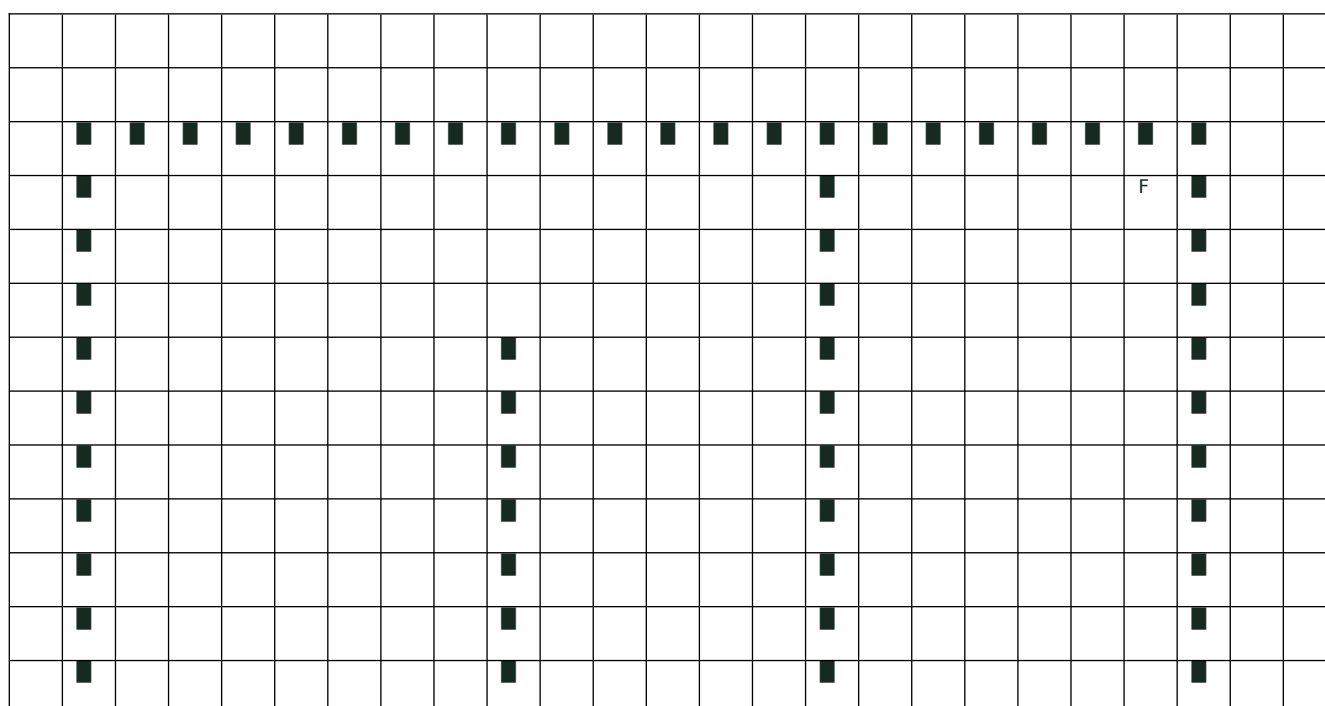
Blocks you may use this week:

- say [blank]
- go to x [-180] y [-120]
- wait until <>
- touching [Finish]?
- say [Made it!]
- forever
- if <> then
- say [Made it!] for [2] seconds
- say [Still at Finish] for [2] seconds

The maze

Page 2 is your stage. It is printed at the real size of the Scratch stage. The stage is 480 wide and 360 tall. x runs from -240 on the left to 240 on the right. y runs from -180 at the bottom to 180 at the top. Grid lines are every 10, so one arrow block moves your counter one square. The six dark blocks are the Maze walls. They are printed at exactly these places: wall along the bottom from x -220 to 220 and y -160 to -140; wall along the top from x -220 to 220 and y 140 to 160; wall down the left from x -220 to -200 and y -140 to 140; wall down the right from x 200 to 220 and y -140 to 140; inner wall 1 from x -70 to -50 and y -140 to 80; inner wall 2 from x 50 to 70 and y -80 to 140. The yellow square is Finish. It runs from x 168 to 192 and y 108 to 132. The start square is marked at x -180, y -120. TOUCHING RULE FOR PAPER: your counter is touching a shape when the middle of your counter sits on that shape, or on its edge. This rule was checked against the lesson's own touching test at every square on this grid. The two agree at every square, so your paper answer and a screen answer are the same answer. WHAT YOU DO THIS WEEK: 1. Put your counter on the start square, x -180, y -120. 2. Read your own win script from grid 1, one block at a time. 3. At say [blank], turn your message card face down. There is no message now. 4. At go to x y, put your counter on your safe start. 5. At wait until, STOP. Do not say your win message yet. The script is waiting. 6. Now use your arrow scripts. Each arrow block moves your counter one square. One square is 10. 7. After each move, write x and y in the trace table. Answer: touching Finish, yes or no? 8. The first time you answer yes, the wait until is finished. Say your win message once. Write it on your message card and turn it face up. 9. Now restart. Turn your message card face down. Put your counter back on the start square. 10. Test a wall. Put your counter on x -60, y 0. Your wall rule sends it back to x -180, y -120. 11. Win a second time. 12. Move only one square at a time. Do not lift the counter over a wall. If your counter lands on a wall, your wall rule sends it back to the start square.

The maze below is the real one, at its real size. One square is 20 steps. Player starts at (-180, -120). Finish is at (180, 120). Move a counter square by square and write down where it lands.



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Predict, trace, compare

Table 1. Predict first. Put your counter on each square. Do not move it yet. Answer before you check.

Place on the maze grid	x	y	Touching Finish? yes or no	Touching Maze? yes or no	What my win script does here
Safe start	-180	-120			
Clear space test	0	0			
Wall 1 test	-60	0			
Wall 2 test	60	0			
Finish test	180	120			

Table 2. Trace your win script. One row for each block you run. Row 1 is filled in for you. Your counter starts on the safe start and your message card is face down.

Step	Block I run	x after this block	y after this block	Message showing now
1	when green flag clicked	-180	-120	none
2	say [blank]			
3	go to x [-180] y [-120]			
4	wait until <touching [Finish]?> — before I move			
5	I move my counter to Finish — see Table 3			
6	say [Made it!]			
7	no more blocks — the win script has ended			

Table 3. Trace your moves into Finish. One row for each arrow block you run. The first row is filled in for you. Put your counter on x 180, y 60 to begin this part.

Move	Block I run	x after	y after	Touching	Does say
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				Finish? yes or no	[Made it!] run yet? yes or no
start	counter placed here	180	60	No	No
1	when [up arrow] key pressed, change y by 10				
2	when [up arrow] key pressed, change y by 10				
3	when [up arrow] key pressed, change y by 10				
4	when [up arrow] key pressed, change y by 10				
5	when [up arrow] key pressed, change y by 10				
6	when [up arrow] key pressed, change y by 10				

Table 4. Compare two win actions. A partner or an adult follows each one exactly as it is written. You watch and count.

Win action	First time my counter reaches Finish	I check again while my counter stays on Finish	How many messages in one run
wait until <touching [Finish]? > then one say [Made it!]			
forever, if <touching [Finish]?			

> then two say for 2 seconds blocks			
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Table 5. Our restart check. Give this page to a partner or an adult. They follow your written script exactly. They do only what is written. You watch. You do not tell them what you meant.

Shared test	We expect	We see
Fresh flag, away from Finish	No win message	
Reach Finish	Win message appears once	
Stay at Finish for another check	No repeated win response	
Green flag after a win		
Touch a wall after the restart		
Reach Finish a second time		

Table 6. Challenge only. My repair investigation.

What I changed	My prediction	What happened when we ran it on paper
I took out forever and if <> then. I put in wait until <touching [Finish]?> and one say [Made it!].		

My completion record

What this week shows

AQA 71638 outcome 4: Observe pupil construction and demonstration of the win script. Record a win and a restarted run. (Teacher guide, Individual evidence.) Audit outcomes 1-4 individually. Do not claim outcomes 5-6 from the teacher model.

- ☐ I created a wait until condition that checks touching Finish.
- ☐ My win message appears once per green-flag run.
- ☐ The flag clears the message and returns Player to the start.
- ☐ The controls and wall rule still work, and I can win again after restart.
- ☐ I wrote the win script in grid 1 myself.
- ☐ My safe start is x _____ y _____. My win message is _____.
- ☐ I moved my counter and I filled in Table 3.
- ☐ Someone else followed my written script in Table 5. I watched them.
- ☐ My idea, and what it changed: _____.
- ☐ Tick, point to, or say each line you agree with. An adult can write your words.

Adult record

- ☐ Construction, observed directly: the pupil chose wait until from the word bank, not forever, and wrote touching [Finish]? inside the hexagon in grid 1 by their own hand.
- ☐ Construction, observed directly: the pupil joined one say block below wait until, and left the first say block empty.
- ☐ Construction, observed directly: the go to block in grid 1 uses the same x and y as the pupil's start script in grid 7.
- ☐ Demonstration, observed directly: the pupil traced their own script in Table 2 and moved their counter square by square in Table 3, and did not say the win message until the first square where touching Finish was true.
- ☐ Demonstration, observed directly: exactly one win message per run. The pupil did not repeat it while the counter stayed on Finish.
- ☐ Demonstration, observed directly: on the green flag the pupil turned the message card face down and returned the counter to the safe start, in that order.
- ☐ Demonstration, observed directly: after the restart the pupil placed the counter on x -60 y 0, applied their own wall rule and returned the counter to x -180 y -120, then reached Finish a second time and gave one message.
- ☐ Independent execution, observed directly: a partner or adult followed the pupil's written script for Table 5 without being told what the pupil meant. Record whether the script needed any spoken clarification before it could be followed.
- ☐ Record the date and the support used against each of the four pupil checks, in the Individual evidence table of this teacher guide.
- ☐ Record here that this pupil took the PAPER ROUTE for Week 6, and that no W06_Win.sb3 exists. Week 7 must reprint this pupil's grid 1 page instead of opening a file.

- ☐ Record completed outcomes on the AQA summary sheet. Write nothing on the AQA summary sheet about which route the pupil took. The route is recorded in this teacher guide only.
- ☐ Do not infer creation from the printed listings on the challenge page. Only grid 1, written in the lesson, evidences creation.

Adult name: _____ Date: _____

Route taken: paper · Recorded in the teacher guide, not on the AQA form.